Bayes Decision Theory & Bayesian Learning

Overview

- Bayes Rule : A review
- Bayesian Decision Theory
  - Loss Functions and Risk
  - Optimal Bayes Classifier
- Naïve Bayes Classifier
- Parameter estimation
  - Maximum Likelihood Estimation (MLE)
  - Bayesian Parameter Estimation
- Examples and Comparisons

Review of Bayes Theorem

Discrete case : If an event B can occur if and only if one of the mutually exclusive events $A_1, A_2, \ldots, A_k$ occurs, then

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)} = \frac{P(B|A_i)P(A_i)}{\sum_{i=1}^{k} P(B|A_i)P(A_i)}$$

Bayes theorem relates aposteriori probabilities with conditional and prior probabilities.

An example

A certain disease affects 2% of the population. A diagnostic test for the disease has an accuracy rate of 95% (i.e., the probability of testing positive given that the disease is present is 0.95), and the false positive rate is 3% (i.e., the probability of testing positive given that the disease is absent is 0.03). If a randomly selected person tests positive, what is the probability that they actually have the disease?

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Solution : Let us define the events below:

- A: A person has the disease.
- B: A person tests positive

Let $\bar{A}$ and $\bar{B}$ represent the events that the randomly selected person does not have the disease and a randomly selected person tests negative for the disease, respectively.

We have from the given data:

- P(A) =0.02 (The probability of having the disease) and hence $P(\bar{A}) = 1 - P(A) = 0.98$
- $P(B|A) = 0.95$ (The probability of testing positive given that the disease is present)
- $P(B|\bar{A}) = 0.03$ (The probability of testing positive given that the disease is absent)

Now, we need to find $P(A|B)$ i.e. the probability of having the disease given that the person tests positive.

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Now, we need to find $P(A|B)$ i.e. the probability of having the disease given that the person tests positive. Using Bayes Theorem we have,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\bar{A})P(\bar{A})}$$

$$\implies P(A|B) = \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.03 \times 0.98)}$$

$$\implies P(A|B) = 0.019/(0.019 + 0.0294) = 0.3926$$

Bayesian Decision Theory

- Framework for decision making under uncertainty
- Based on probabilities and costs (loss)
- Goal: minimize expected loss

Key Components:

- Prior: $P(\theta)$
- Likelihood: $P(D|\theta)$
- Posterior: $P(\theta|D)$

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Loss Function and Risk

Loss Function $L(\theta, a)$: Measures penalty of taking action a when true parameter is θ

Expected Risk:

$$R(a) = \sum_{\theta} L(\theta, a) P(\theta | D)$$

Optimal Decision Rule: Select the action a^* which minimizes the risk i.e.

$$a^* = \arg\min_{a} R(a)$$

Common Loss Functions

- 0-1 Loss:

$$L(\theta, a) = \begin{cases} 0 & \text{if } a = \theta \\ 1 & \text{otherwise} \end{cases}$$

- Squared Error Loss:

$$L(\theta, a) = (\theta - a)^2$$

Result:

- 0-1 loss $\Rightarrow$ MAP estimate
- Squared loss $\Rightarrow$ Posterior mean

Optimal Bayes Classifier

The optimal Bayes classification criterion (**optimal because it minimizes classification error**) is to assign the class label that corresponds to the highest posterior probability i.e. $y^* = \arg\max_y P(y|\mathbf{x})$

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Using Bayes rule: $P(y|\mathbf{x}) = \frac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x})}$.

Hence the decision rule can be written as $y^* = \arg\max_y P(\mathbf{x}|y)P(y)$

Bayes Rule in classification

Let $C_1, C_2, \ldots, C_K$ be the class labels. Let $\mathbf{x}$ be the feature vector representing an instance which needs to be classified. Let y be the predicted class label.

$$P(Class = C_i|\mathbf{x}) = \frac{P(\mathbf{x}|C_i)P(C_i)}{P(\mathbf{x})} \text{ where,}$$

- $P(Class = C_i|\mathbf{x})$ is the aposteriori probability
- $P(C_i)$ is the prior probability of class C_i
- $P(\mathbf{x})$ is referred to as the evidence.

Decision Rule

Compute posterior for each class and assign the class label that corresponds to the highest aposteriori probability.

An example using the iris Dataset

- **3 classes:** Setosa, Versicolor, Virginica
- **Features:** Sepal length (cm), Sepal width (cm), Petal length (cm), Petal width (cm)

Sepal LengthCm	Sepal WidthCm	Petal LengthCm	Petal WidthCm	Species
5.1	3.5	1.4	0.2	Setosa
6.4	3.2	4.5	1.5	Versicolor
6.4	2.8	5.6	2.1	Virginica

Table: A few selected records from the iris dataset

An example using the iris Dataset

Given a new instance SepalLength $=6.3$ Cm,SepalWidth $=3.3$ Cm,PetalLength $=4.7$ Cm,PetalWidth $=1.6$ Cm predict the class.

- Let $\mathbf{x}$ be the vector representing the given instance. Then $\mathbf{x} = [6.3, 3.3, 4.7, 1.6]^T$
- Let Setosa = class C_1, Versicolor = class C_2 and Virginica = class C_3. We need to assign the class label corresponding to the highest aposteriori probability among $P(Class = C_1|\mathbf{x}), P(Class = C_2|\mathbf{x}), P(Class = C_3|\mathbf{x})$

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An example using the iris Dataset

- The prior probabilities are easily estimated from the data. The iris dataset consists of 50 instances each of Setosa, Versicolor and Virginica respectively. Hence, $\mathrm{P}(\mathrm{Class} = C_1) = \mathrm{P}(\mathrm{Class} = C_2) = \mathrm{P}(\mathrm{Class} = C_3) = 1/3$
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- Need to compute **class conditional densities** $p(\mathbf{x}|Setosa), p(\mathbf{x}|Versicolor), p(\mathbf{x}|Virginica)$ Usually a specific underlying probability density/distribution is assumed and the parameters of the distribution are estimated from the training data. This is the **Parameter estimation** problem.

An example using the iris Dataset

Suppose we model the class conditional densities as multivariate Gaussians. Let $(\mu_1, \Sigma_1), (\mu_2, \Sigma_2), (\mu_3, \Sigma_3)$ be the mean vector and covariance matrices corresponding to the classes C_1, C_2 and C_3 respectively. The class conditional density for class C_i $i = 1, 2, 3$ is given by

$$p(\mathbf{x}|C_i) = \frac{1}{(2\pi)^{d/2}|\Sigma_i|^{1/2}} exp(-\frac{1}{2}(\mathbf{x} - \mu_{\mathbf{i}})^T \Sigma_i^{-1}(\mathbf{x} - \mu_{\mathbf{i}})) \tag{1}$$

Class (i)	μ_i	Σ_i
C_1 = Setosa	[5.006, 3.428, 1.462, 0.246]'	
C_2=Versicolor	[5.936, 2.77, 4.26, 1.326]'	
C_3=Virginica	[6.588, 2.974, 5.552, 2.026]'	

Probabilistic Generative Classifiers

Model joint distribution:

$$P(x, y) = P(x|y)P(y)$$

Steps:

- Learn $P(y)$ (class prior)
- Learn $P(x|y)$ (class-conditional)
- Use Bayes rule to classify

Examples:

- Naïve Bayes
- Gaussian Discriminant Analysis

Naive Bayes

- A simple and efficient classification technqiue which often performs well in practical applications involving large amounts of data, example include text classification (e.g., spam detection) and sentiment analysis.
- "Naive assumption" : Features are **conditionally independent given the class label**

Naive Bayes Classifier

Assumption of conditional independence:
Suppose $\mathbf{x} = (x_1, x_2, ..., x_n)$ is the feature vector and y is the class label.
Then conditional independence implies

$$P(x_1, x_2, ..., x_n|y) = \prod_{i=1}^{n} P(x_i|y)$$

Accordingly, the optimal Bayes classifier is :
Assign the class label $y*$ to instance $\mathbf{x} = (x_1, x_2, ..., x_n)$ where,

$$y^* = \arg\max_y P(y) \prod_{i=1}^{n} P(x_i|y)$$

Types of Naive Bayes

- Gaussian Naïve Bayes
- Multinomial Naïve Bayes
- Bernoulli Naïve Bayes

Choice depends on feature distribution.

Naive Bayes on Iris

Given feature vector $\mathbf{x} = (x_1, x_2, x_3, x_4)$
compute $P(C \mid X) \propto P(C) \prod_{i=1}^{4} P(x_i \mid C)$

Naive Bayes on Iris

Assuming Gaussian distribution for features,
For each feature:

$$P(x_i \mid C_i) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right)$$

- μ_i: mean of feature for class C
- σ_i^2: variance

Example Calculation with only two features

Suppose:

- $P(Setosa) = 0.33$
- $P(x_1 \mid Setosa) = 0.8$
- $P(x_2 \mid Setosa) = 0.7$

Then:

$$P(Setosa \mid X) \propto 0.33 \times 0.8 \times 0.7$$

$$= 0.1848$$

Parameter estimation

Bayes Rule for parameter estimation

Let D denote observations (data) and θ is the set of parameters that need to be estimated from the observed data. Application of Bayes Rule yields

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

- $P(\theta|D)$: Posterior
- $P(D|\theta)$: Likelihood
- $P(\theta)$: Prior
- $P(D)$: Evidence

$$P(D) = \int P(D|\theta)P(\theta)d\theta$$

Maximum Likelihood Estimation (MLE)

Definition : $\hat{\theta}_{MLE} = \arg\max_{\theta} P(D|\theta)$

- MLE estimates parameters by maximizing likelihood
- Ignores prior information

It is often convenient to maximize the **log-likelihood**

$$\ell(\theta) = \log P(D|\theta)$$

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For i.i.d data : $P(D|\theta) = \prod_{i=1}^{n} P(x_i|\theta)$
and log-likelihood is :

$$\ell(\theta) = \sum_{i=1}^{n} \log P(x_i|\theta)$$

Maximum A Posteriori (MAP)

Maximum A Posteriori (MAP): $\theta_{MAP} = \arg\max_\theta P(\theta|D)$
Using Bayes Rule:

$$\theta_{MAP} = \arg\max_\theta P(D|\theta)P(\theta)$$

Log form:

$$\log P(D|\theta) + \log P(\theta)$$

Interpretation : MAP = MLE with prior regularization
MAP incorporates prior knowledge:

MLE vs MAP

- MLE ignores prior
- MAP includes prior

$$\text{MLE: } \theta_{MLE} = \arg\max P(D|\theta)$$

$$\text{MAP: } \theta_{MAP} = \arg\max P(D|\theta)P(\theta)$$

- With uniform prior: MAP = MLE

MLE Example (Bernoulli)

Data: $D = \{x_1, x_2, \ldots, x_n\}$, $x_i \in \{0, 1\}$

$$P(D|\theta) = \theta^{\sum x_i}(1-\theta)^{n-\sum x_i}$$

Log-Likelihood:

$$\ell(\theta) = (\sum x_i)\log\theta + (n - \sum x_i)\log(1-\theta)$$

MLE Solution:

$$\hat{\theta}_{MLE} = \frac{1}{n}\sum x_i$$

Bayesian Parameter Estimation

Goal: Compute posterior distribution

$$P(\theta|D) \propto P(D|\theta)P(\theta)$$

- Incorporates prior knowledge
- Output is a distribution, not a single value

MLE vs Bayesian

- MLE:
  - Uses only data
  - Can overfit with small data
- Bayesian:
  - Uses prior + data
  - More robust with limited data

Key Takeaways

- Bayes Rule updates beliefs
- MLE ignores prior
- MAP incorporates prior knowledge
- Optimal Bayes classifier minimizes error
- Naïve Bayes assumes independence
- Generative models learn data distribution
- Bayesian regression captures uncertainty

Summary

- Bayesian decision theory minimizes expected loss
- MLE maximizes likelihood
- MAP incorporates prior into estimation
- Bayesian methods provide full uncertainty modeling